NETFLIX - ROOT CAUSE ANALYSIS

Problem Statement : You are the product manager at Netflix. You see there is a 10% drop in the Average View Duration. Find the Root Cause.

Defining Netflix?

Netflix, founded by Reed Hastings and Marc Randolph in 1997, started as a DVD rental service and later transformed into the world's leading subscription-based streaming platform.

It provides an extensive library of films, TV shows, documentaries, and exclusive Netflix Originals. The company operates in over 190 countries, offering localized content in multiple languages.

Netflix's unique recommendation algorithm personalizes content for each user, enhancing their viewing experience.

Over the years, it has reshaped the entertainment industry and maintained a strong influence in the global media landscape.

1. Clarifying Questions (To understand the context better)

1.1 User Segmentation:

1.1.1 Is the drop uniform across all user segments (regions, age groups, device types)? Yes.

1.1.2 Are new or returning users more affected?
All of them .

1.1.3 Is there a specific geographic region where the drop is more pronounced?
No.

1.2 Content-Specific:

1.2.1 Is the drop specific to certain genres, shows, or movies?
All genres.

1.2.2 Are new releases underperforming compared to expectations?
Yes.

1.2.3 Did we change content recommendations recently?
No

1.3 Time Period:

1.3.1 When did we first notice the drop?
Approximately two weeks.

1.3.2 Is it seasonal (e.g., holidays, sports events affecting viewership)?
No such big event.

1.3.3 Are there external factors (e.g., major events, competitors' new releases)?
There are a few releases but not major events.

1.4 Platform/Device:

1.4.1 Is the drop consistent across devices (mobile, smart TV, web)?
Yes , All platforms

1.4.2 Have there been any recent UI/UX changes on specific platforms?
No such changes.

1.5 User Behavior:

1.5.1 Are users abandoning content at specific timestamps?
No.

1.5.2 Is there an increase in content switching or exits?
No.

1.5.3 Have there been changes in binge-watching behavior?
No.

1.6 Definition

1.6.1 How the AVD is calculated is it time duration or one active view per user? **the definition of average view duration here (for e.g. if a user watches the same content multiple times in a single session, would each play be counted separately for Average View Duration, or is it treated as a single play?**

Each play would be counted separately for Average View Duration, so if a user watches the same content multiple times, those would each count as individual plays in the calculation

1.6.2 10% drop as calculating all users view taking into consideration or calculating average view per customer.
Yes , we are considering the views across the generes.

1.6.3

2. Sanity Check Questions (To rule out internal issues or anomalies)

2.1Technical Issues:

2.1.1 Are there app crashes, buffering issues, or performance slowdowns?
2.1.2 Any recent infrastructure changes that could impact streaming quality?
2.1.3 Have CDN issues affected content loading times?

2.2 A/B Testing:
2.2.1 Were there any experiments or A/B tests deployed recently?
2.2.2 Are test groups experiencing different view duration patterns?

2.3 UI/UX Changes:
2.3.1 Have there been any layout, autoplay, or navigation changes that could impact engagement?
2.3.2 Has the recommendation algorithm changed recently?

2.4 Pricing and Subscription:
2.4.1 Have there been changes in subscription plans, promotions, or price hikes that may affect viewing patterns?
2.4.2 Are users canceling or downgrading their plans more frequently?

2.5 Content Availability:
2.5.1 Have any popular titles been removed recently?
2.5.2 Are there licensing issues causing unexpected content unavailability?

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3. Pivot Questions (If the initial hypotheses don't reveal the root cause)

3.1 Competitor Landscape:
3.1.1 Are competitors (Disney+, Prime Video, etc.) launching compelling new content?
3.1.2 Are users spending more time on alternative platforms?

3.2 Social Media Trends:
3.2.1 Are negative reviews or social media feedback influencing user sentiment?
3.2.2 Is there a viral trend affecting viewing habits?

3.3 Market Trends:
3.3.1 Are there industry-wide changes (e.g., shorter attention spans, shift to short-form content like YouTube Shorts)?

3.4 User Surveys/Feedback:
3.4.1 Have there been complaints or concerns raised in customer support channels?
3.4.2 Are there qualitative insights indicating dissatisfaction with content quality?

4. **External and internal factors -**

4.1.1 Is there any Political events like elections or any political events are going on due to that any restriction imposed?

- No

4.1.2 is there any new entry of the similar product or competitor or something exclusive offering they are providing?

- Not any major competitor released but there was a business deal between Jio cinema of reliance industries in india and Disney+ hotstar. any kind of backlash facing on social media like on Twitter or Facebook due to any change in policy?

- No

4.1.3. Is there any recent regulatory announcement or policy affecting the industry? (Example: Stricter data privacy laws like GDPR enforcement in Europe)

- No

4.1.4 Are there any ongoing cultural or national holidays that could be diverting audiences towards outdoor or family activities? (Example: Thanksgiving in the U.S. or Chinese New Year)

- No

4.1.5 Any environmental factor? As I heard about the recent Los Angeles wildfires that devastated multiple homes in the Western United States—could that have contributed to the drop in average view duration?

- Yes, there was a few the wildfires in the Western United States and typhoon in south east asia.

5. Internal

5.1.1 Is there any changes in the content repository?

Yes some very popular web-series are now not in the Netflix

5.1.2 Is there new content added?

Yes, there are various new contents added

5.1.3 What is the customer feedback of the new web series or content?

Most of the newly added web series are not getting good feedback fom the viewers.

5.1.4 Is there any changes of the average time spent on various pages?

Yes , viewers are spending more time in the landing page and with the search options.

5.1.5 Is there any issue reported as customers are having issue with the content on the page?

Yes there are multiple issues reported across the world i.e. viewers are having issue finding their view like content.

User journey analysis-

1. **User opens the Netflix app / Web app / Web site**
2. **User specific content with recommendation and last watched view.**
3. **Browsing for other content /newly added content**
4. **Selecting the content and play**
5. **Feedback of the content –**
6.

Root cause Hypothesis-

Proposed solutions-

Short term solutions

Long term Solutions-

Clarifying Questions?

- **When did the drop begin?**

  —> It was noticed 2-3 weeks ago. Though it was not a sudden shift, it gradually decreased over 2 weeks.

- **Is the drop global or regional?** —> Globally
- **Is the drop across all devices or specific ones (e.g., mobile, TV)?**
- `—> All Devices`
- **Are all content categories impacted (Web series or Movies), or is it isolated to specific genres?**
- `—> All content categories impacted`
- **Is completion rates for episodes have dropped or users dropping off earlier?**
- `—>Yes, completion rates for episodes have dropped, especially in`

the middle of the series.

- **Has AVD dropped across all streaming platforms, indicating an industry-wide shift?**
- `—> No, based on recent trends, it appears that AVD has dropped only on Netflix.`
- **Are there seasonal patterns (e.g., holidays, exams, major sports events) impacting user habits?** —> Yes, this period coincides with significant seasonal shifts—holidays and major sports events like the NFL season or the FIFA World Cup.
- **Has the economic landscape shifted, affecting users' ability to maintain subscriptions?** —> No
- **Have competitors launched aggressive campaigns?** —> Yes, platforms like Disney+ and Amazon Prime Video ran strong marketing campaigns, attracting viewers.
- **Is there a competing platform that recently released major titles or features?** —> Yes, Disney+ and Amazon Prime released major movie ad series, which likely contributed to the shift in viewer attention.
- **When did the competitor launch these major series?**
- `—> 2 weeks ago`
- **How does the timeline of their launch correlate with the drop in AVD?** —>The AVD drop coincided with the release dates of these series, suggesting they diverted significant viewership from Netflix
- **Any examining changes Netflix made, such as content releases?** —>Yes there are delays in high-demand content releases due to some conflict between the production team, which affected new releases in those genres.

- **Is their any Recent changes or updates in the UI/UX, recommendation algorithms, or settings** that could disrupt user flow. —> No such recent updates has been done.
- **Are users facing issues like buffering, app crashes, or slow load times?** —> Some users have reported occasional buffering and slow load times, especially in regions with weaker internet connections.
- **Are there complaints or comments from users mentioning dissatisfaction with the content lineup?** —> Yes, user feedback suggests dissatisfaction with recent content quality and the slow release of high-engagement titles.

Identified Root Cause:

Based on initial analysis, one of the possible causes seems to be **a lack of high-engagement content** during this period, compounded by :

- **Delayed New Releases:** Production setbacks have resulted in fewer new and highly anticipated original titles being released.
- **Content Quality Concerns:** Survey results show that users are dissatisfied with the quality and engagement of recent content.
- **Increased Competition:** Competing platforms like Disney+ and Amazon Prime Video launched popular content around the same time, which may have diverted viewers from Netflix.

Addressing The Root Cause :

1. **Strengthen the Content Pipeline:**
    - Accelerate the release of popular, high-demand titles in the short term.
    - Diversify the catalog to include trending genres and formats.
    - Use data to identify underperforming content categories and adjust accordingly.
2. **Respond to Competitive Pressure:**
    - Launch aggressive marketing campaigns to highlight Netflix Originals and upcoming releases.
    - Offer incentives like sneak peeks or early access to premium content for loyal subscribers.
    - Differentiate Netflix by emphasizing unique features like superior personalization.
3. **Enhance User Experience:**
    - Address any technical issues that might disrupt viewing.
    - Optimize recommendation algorithms to surface highly relevant content for users.

Metrics to be tracked:

- **Average View Duration Recovery:** Monitor if AVD trends upward after implementing solutions.
- **Content Performance Metrics:** Analyze completion rates, ratings, and engagement for new releases. Check if the **Net Promoter Score** has increased
- **Retention Rates:** Check if Average watch time has increased.

- **User Sentiment:** Look for improvements in surveys , app reviews, and social media mentions.